Hierarchical Feature Selection Framework over U.S. Census Data

By: Viraat Udar

Advisor: Dr. Kanika Sood

Table of Contents

Introduction....3
Background....3
Objectives....5
Statistical & ML Background....7
Pearson & Spearman Correlation....7
Benjamini-Hochberg False Discovery Rate Correction....8
CatBoost: ordered boosting & oblivious trees....9
K-fold cross-validation....12
Definition of the Problem....13
ACS Data Infrastructure....14
ACS 5-year estimates product....14
DuckDB architecture & KV cache....15
Vintage stability & variable drift....16
Statistical Screening Layer....16
Dataset assembly....16
Univariate screening procedure....17
Target datasets & loaders....18
Recipe Sampler & ML Layer....19
Recipe sampling design....19
CatBoost evaluation per recipe....19
Score aggregation: column, family, pairwise synergy....20
Final model training....21
Pipeline Architecture....21
Cross-dataset Comparison....23
Problems & Possible Improvements....23
Bibliography....25

Introduction

Most real-world datasets can only provide a portion of the story they could tell, no matter the subject of the data. The missing portion often is the context of the data, and the relationship of that context to the data itself. For datasets that involve human lives in some way, the outcomes are shaped by countless human factors. Relative to the original data this contextual data is dramatically larger in size, making the challenge in telling this story about identifying the contextual data that is most relevant. To ground this challenge in a concrete setting, we turn to a true source of contextual information that the United States government and public use everyday, the United States Census [1]. We will dive into how the Census data is collected and privacy protections to clarify the opportunities it offers and the constraints it imposes on analysis.

Background

The United States Census dates back to before the founding of our country. Originally the practice was adopted from the British, who has been taking a census in almost all of their colonies. During the founding of the United States, the Founding Fathers included the Census as a constitutional mandate. At the time, the Census was a simple population count, used to apportion seats in the House of Representatives [2]. In today's landscape, the Census is now more comprehensive and is used to determine funding for schools, housing, road, and emergency services.

Though the Census collects a vast number of key data metrics, the information isn't immediately available to the public. The Census Act of 1879, was crucial legislation for a number of reasons, but very importantly it made into law that the census would now include privacy protections [4]. When the US Census was first

conducted the US Marshal would post the results to the public, in town squares. 89 years later, the 72-Year Rule would be passed, putting into law that detailed census records would become public only after 72 years [3]. The reasoning was simple, the Census Bureau wanted to build public trust, so that it could ensure participation. When the Census would be released to the public, there was a growing fear of taxation and inquiry, which began leading to a low response rate. For today, the Census data of individuals that is released to the public is for 1950. This is great for historians, but not for data analysis. However aggregated data from the Census, is released as soon as they are available. Currently computerized aggregate data describing the characteristic of a small geographic area, is available for the entire period from 1790-2010.

The Census Bureau conducts two distinct data collection programs relevant to this work. The Decennial Census, taken every ten years, provides a complete population count but is limited in the depth of information it captures. The American Community Survey (ACS) is an ongoing annual survey that collects detailed social, economic, housing, and demographic characteristics continuously throughout the decade. Because the ACS samples the population on a rolling basis, the Bureau publishes a 5 year estimate product that pools five consecutive years of survey responses into a single dataset. At the state and county level, where individual year samples may be too small to produce reliable estimates, this pooling substantially improves statistical stability. For analyses involving chronic disease incidence or long term economic trends, the 5 year window also captures the sustained demographic conditions that shaped the outcome rather than a single year snapshot. To this end, the primary analyses in this project use a

5 year aggregation window for target datasets wherever the multi-year pooling is best suited. This will match the temporal scope of the ACS survey. In cases where signals may be stronger under single-year conditions, a 1 year window can be retained.

The census data is a great tool for a data center, but it comes with its challenges for standard data analysis. With so many possible contextual features to draw relationships with, a robust method of untangling redundant but relevant columns is needed. For the purpose of clarity, it can be helpful to keep in mind some example target datasets that can be used with this system. To ground this in concrete terms, consider three example target datasets used to validate this system: opioid overdose mortality rates by state, lung and bronchus cancer incidence by state, and UFO sightings by state. Each represents a different relationship between the target data and the demographic context the Census provides. The system presented will seek to identify the best features and feature families to predict these values per state or county.

Objectives

Existing approaches to data analysis at this scale rely on manual feature selection and correlation methods that cannot account for the sheer size of contextual relationships of such large datasets as the U.S. Census. Such methods include correlation matrices and principal component analysis (PCA), both of which assume linear contextual relationships and independence among variables [5]. In the case of Census data, neither of these assumptions are the case requiring a more robust approach to discovering the non-linear contextual relationships and overlapping hierarchical relationships among features. In order to extract meaningful contextual features, the analysis system must be able to sample, score, and

group combinations of features by parsing existing hierarchical connections, rather than considering features in isolation. The term ‘features’ will be used to refer to Census data columns like ‘Percent of population with employment’. The term ‘data points’ will refer to values belonging to a state or county. Lastly the term ‘target data’ will be used to refer to the user’s dataset, as this may change between executions.

Despite meaningful connections existing among feature and target data, the structure of Census analysis introduces two compounding sparsity problems. The first is geographic, where target data rarely exists for every state or county, as collection methods differ between states and some values are suppressed entirely. The system deals with this but operating on partial datasets with one datapoint per state or county. The next problem is more dimensional, due to the ACS feature matrix providing approximately 528 datapoints per geography where state-level data provides at most 51 rows.For traditional machine learning methods, the ratio of datapoints to rows is too high, which would result in false patterns even if a real signal exists. This system will require a screening stage prior to any model training, this will reduce the feature space to a smaller candidate set where a real signal can be distinguished from statistical noise.

The analysis approach for these problems will use a three-layer pipeline. Due to the high ratio of datapoints to rows, neural network based approaches were ruled out early on. State level data contains only 51 rows or less depending on target data availability, but county level data can be even more scarce with Delaware for example which only has 3 counties. Modern neural networks require a large amount of rows, orders of magnitude above what would be available through the ACS data, therefore they are unsuitable for these analyses unlike machine-learning methods which are not as needy. In order to address the sparsity problem, the two layers are as follows: The first layer is for statistical screening, this will reduce the 528 column feature space to a smaller set of candidates which have some association to the

target data with either Pearson's or Spearman’s correlation. The second layer will use CatBoost, a machine learning model which will learn non-linear relationships within the reduced feature set. The third and final layer is the hierarchical feature family selection, which samples and scores combinations of features based on the ACS family structure that they belong to. Through accumulation of CatBoost training results, the most relevant and interpretable set of features will converge from the repeated iterations.

Statistical & ML Background

Pearson & Spearman Correlation

The first layer of the pipeline reduces the ACS features matrix from approximately 528 columns to a candidate set using the association between every feature and the target variable independently. We use two correlation coefficients for this, Pearson's r and Spearman’s ρ.

Pearson’s r measures the strength and direction of the linear association between two variables and is defined as follows [8]:

$$r = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2 \sum_{i=1}^{n} (y_i - \bar{y})^2}}$$

Spearman’s ρ assumes a monotonic relationship between ranked variables and is essentially a Pearson correlation using variable ranks rather than variable values [9]. It is defined as:

$$rho = 1 - \frac{6 \sum_{i=1}^{n} d_i^2}{n(n^2 - 1)}$$

where d_i is the difference between the rank of x_i and the rank of y_i.

For each of 528 E-suffix ACS columns, both coefficients are computed towards the target variable. The count data in ACS is heavily right-skewed, whereby variables like total

employment or household counts span several orders of magnitude making Spearman a strong measure in most cases.We maintain Pearson's correlation here because it provides power for features that have nearly linear relationships to the target.

Benjamini-Hochberg False Discovery Rate Correction

A multiple testing problem arises by computing the correlations of 528 features simultaneously. With a fixed significance threshold of $p < 0.05$, about 5% of features could register as significant just by random chance rather than a genuine finding.

The Benjamini–Hochberg procedure was introduced as “a practical and powerful approach to multiple testing” that controls the false discovery rate (FDR), defined as the expected proportion of falsely rejected hypotheses among all rejected hypotheses, rather than the familywise error rate [10].

Benjamini and Hochberg show that this criterion is equivalent to the familywise error rate when all null hypotheses are true, but is smaller otherwise, which makes it especially useful in large-scale screening settings where some false positives may be acceptable so long as their proportion is controlled [10].

Their procedure orders the p-values from smallest to largest and rejects hypotheses up to the largest rank k for which:

$$p_{(k)} \leq \frac{k}{m} \cdot q$$

where m is the total number of tests and α is the chosen FDR level [10]. The largest k satisfying this inequality determines the cutoff, all features with p-values at or below $p_{(k)}$ are declared significant.

This project applies BH correction to both Pearson and Spearman p-values with $q = 0.1$, so that at most 10% of features that pass are false positives. Thereby, a feature would pass

screening if its adjusted Pearson or Spearman p-value falls under the threshold. By using both adjusted coefficients, fewer real signals will get filtered out before reaching the model. In addition, there is an additional requirement for a minimum absolute correlation above the baseline average of all feature correlations. At n = 51, features like housing and employment can appear statistically significant. This is a population confounding problem where features can appear to correlate due to the influence of high population attributing to target data. Put another way, some ACS features are directly proportional to population size and therefore can appear proportional to target data. Using a threshold of R^2 values, the system filters out weak associations without large configurations to the target dataset. Also by using the existing result data, even with target data that is very weakly correlated the features that are "best" can still filter through.

CatBoost: ordered boosting & oblivious trees

Ordered boosting is a form of gradient boosting that calculates gradients sequentially. This ensures that each datapoint's training corrects for the residual errors of the trees before it.

$$F_m(x) = F_{m-1}(x) + eta \cdot h_m(x)$$

where $F_m(x)$ is the updated ensemble at step m, $F_{m-1}(x)$ is the previous ensemble, eta is the learning rate, and $h_m(x)$ is the new tree added at step m.

This prevents the model from seeing the same data point in both training and validation, eliminating the source of bias that standard gradient boosting is susceptible to at small n. In standard decision trees, branches can be split on different features at each depth level, resulting in an inconsistent structure that changes across runs when the feature set changes. Oblivious decision tree forces the same split condition (feature and threshold) across all branches at any given depth. This results in a tree which is both symmetric and has decision that are directly comparable across runs. This consistency is essential for this project where

the model is retraining repeatedly on different subsets of the features during the recipe sampling phase.

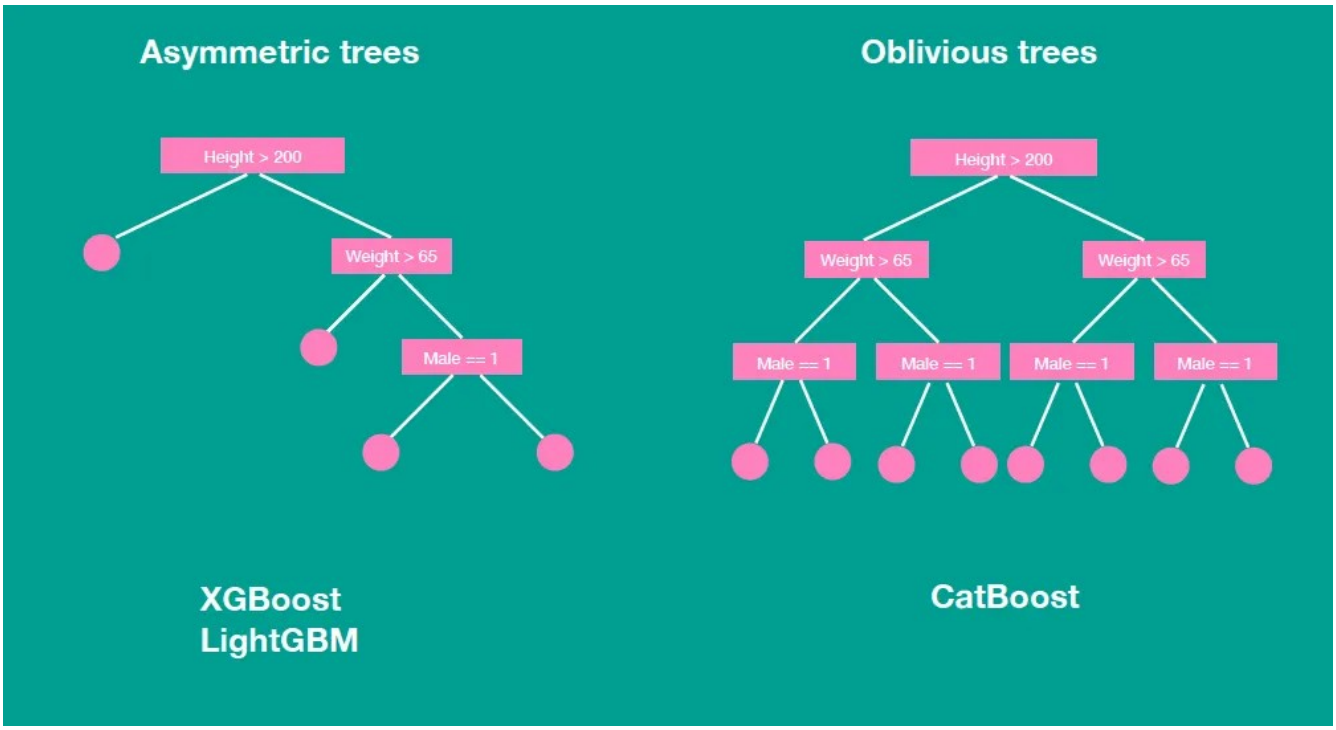


Figure 1: A standard decision tree (left) splits on different features at each node. An oblivious decision tree (right) enforces the same split at every node within a depth level. [7]

K-fold cross-validation

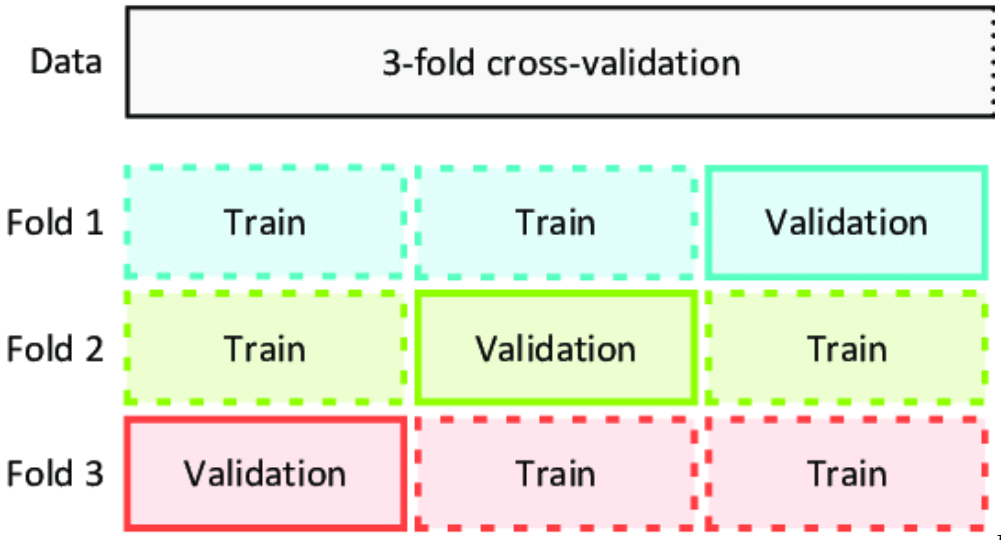


Figure 2: 3-fold Cross-Validation diagram (Will be redrawn from scratch to avoid sourcing) [6]

The mean validation score across all k folds is then reported as the performance estimate:

$$CV_k = \frac{1}{k} \sum_{i=1}^{k} R_i^2$$

where R_i^2 is the score on the i-th held-out fold.

K-fold cross-validation plays 2 roles at different stages of the pipeline. The first role is within the recipe sampling phase, where each recipe of a CatBoost model gets trained and scored using a 3-fold cross-validation scheme. With n = 51 rows, 3-fold gives sizes of 34 examples for training. The intention is not to produce an absolute performance estimate, but rather to have a stable relative ranking across recipes. The variance of standard single split training-validation iterations can lead to misleading results due to the randomness of splitting. To mitigate the results of unlucky splits, the system scores features with respect to stability

across folds. Features that score consistenly across all three folds will have a higher ranking than ones that have the same mean but higher variance in folds.

The second role is the final model evaluation. When the recipe sampling phase converges on a winning feature set, a final CatBoost model is trained and evaluated on that feature set.

$$score_{col} = \frac{1}{3}\sum_{i=1}^{3} R_i^2 - \sqrt{\frac{1}{3}\sum_{i=1}^{3}\left(R_i^2 - \bar{R}^2\right)^2}$$

Definition of the Problem

The sparsity of data compounds with the sheer breadth of ACS features, leading to problematic hurdles that drive every design decision in the project. Availability of target data further restricts to max number of rows available for any number of external factors. The problem of dimensional sparsity manifests within that constraint, as mentioned before 528 candidate features measured against 51 observations would surface coincidental associations rather than genuine ones. The practical solution is to reduce the feature space before any model sees it, which is the role of the statistical screening layer.

A further constraint is that most ACS features are highly collinear. Features like income, poverty, and employment move together when comparing across geographies in a way that makes it difficult to attribute signals individually. The aspect is not a problem that can be solved prior to analysis as it is a structural part of the ACS data. The recipe sampler layer addresses this aspect by never training on the full feature set at once, instead measuring combinations of random features to accumulate stable predictive signals. The system identifies demographic contexts that are associated with outcomes at the geographic level. It does not and cannot identify individual-level causal relationships [11].

ACS Data Infrastructure

ACS 5-year estimates product

The American Community Survey (ACS) publishes its data with 4 Data Profile (DP) groups. DP02 covers social characteristics like education and language; DP03 covers economic characteristics; DP04 covers housing characteristics; and DP05 covers demographic characteristics like age and ethnicity. Across all 4 groups, there are approximately 4224 variables returned for a single geography. The value for a single feature is often provided in several modalities. These are shown in the Table 5.1

Table 5.1: ACS Feature suffixes table

Suffix	Type	Used in analysis
E	Estimate (raw count)	Yes
PE	Percent estimate	Yes
M	Margin of error on E	No
PM	Margin of error on PE	No
EA	Annotation on E	No
MA	Annotation on M	No
PEA	Annotation on PE	No
PMA	Annotation on PM	No

Out of the >4000 variables returned, roughly half are E or PE columns, which are suitable for analysis. The remaining variables serve metadata and annotative purposes and do not provide analytical value. These are filtered out early on in the data collection process. Including both E and PE columns in the same feature matrix would result in near-perfect collinearity between the pair. For this reason, the pipeline filters for only E columns, resulting in approximately 528 usable features for a single geography and year.

DuckDB architecture & KV cache

ACS data has to be made available across all three layers through DuckDB database files, as shown in Figure 5.1. The central cache holds raw JSON blobs keyed by geography, year and variable grouping. A single state database holds all of the state-level ACS rows, while county level rows are stored in separate per-state files.

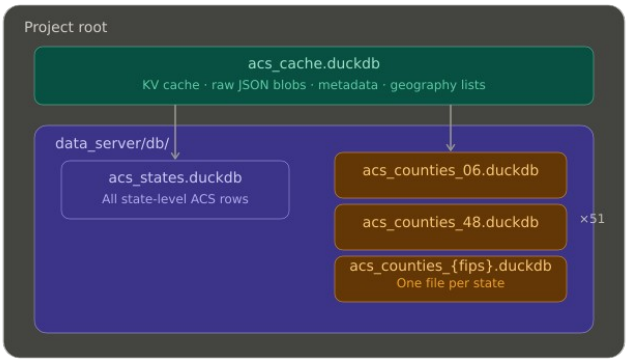


Figure 5.1: DuckDB storage layout. The KV cache at project root feeds both the central states database and the per-state county files.

When a data endpoint is called the server will follow the request flow shown in Figure 5.2. First it will attempt a KV cache lookup. On cache hit, the raw data gets returned immediately without a network call. On a cache miss the server fetches from the US Census API in batches of 45 variables, as well as storing the data in the cache.

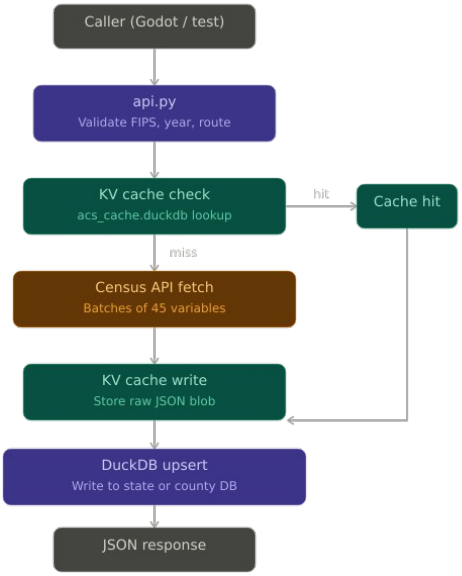


Figure 5.2: Request flow for a data endpoint call. The KV cache is always checked first; the Census API is only contacted on a cache miss.

Vintage stability & variable drift

Because ACS census data collection is legally driven, ACS variables and definitions have changed over time to best reflect the United States. Two major changes are of particular note. The 2019-2020 change involved a major redesign of race and ethnicity questions in DP05. The 2022-2023 change introduced new variables which led to a renumbering of existing variables at the tail of the table. The result of this instability is that variable labels have to be fetched specifically for the data that they describe.

Statistical Screening Layer

Dataset assembly

Prior to screening, the pipeline will create a joint dataset from the ACS feature matrix and the user's target data. The ACS data is read from the DB table filtered for E-suffix columns only and coerced to numeric values. The resulting feature matrix X has approximately 528 columns, with one row for each geography. The user target data is supplied at a variable Y. The ACS data matrix X and user target data column Y are joined

together using the FIPS code. If any datapoint is missing from the user target data, it will be dropped altogether from the prepared dataset.

The prepared dataset can only contain valid rows of ACS and user target data. Based on the target dataset, only available rows of ACS and user target data can remain which would be less than or equal to the total possible amount. For example, the CDC suppresses counts below 10 opioid related deaths to protect the identities of individuals and families, and as a result some states with low rates are absent from the data. None values returned by the Census API are preserved as null rather than zero. This is meaningfully different as zero would create false signals in the feature matrix.

Univariate screening procedure

With the joint dataset now assembled, the screening process computes the association between each column in X with the target variable Y independently. The association consists of the Pearson and Spearman correlation. The BH-FDR procedure is applied to both Pearson and Spearman p-values with a false discovery rate of $q = 0.1$. A feature si considered to pass in this stage if either its adjusted Pearson or Spearman p-value falls under the threshold. An additional requirement is applied regardless of result, whereby each passing feature must also show a minimum absolute correlation above the calculated threshold with the target variable. Statistical significance can come about simply from larger states having more of everything. This would a result of certain features like housing and employment that are directly proportional to population, and target data with similarly but not the same proportionality can be highly correlated.

This guard is particularly important for datasets like cancer incidence where the raw count target data is strongly confounded by population size. The output of the screening procedure is a ranked table of column statistics containing the Pearson and Spearman

correlations and adjusted p-values for every tested feature and passing variable codes. This is the reduced set that will be trained upon.

Target datasets & loaders

Four target datasets are used in this project to demonstrate the pipeline with meaningfully different types of relationships among demographic context and outcome.

Opioid overdose mortality data is sourced from the CDC's WONDER database, and covers all 50 states and the District of Columbia through the years 2018-2024. By defaults, the metric is for age-adjusted death rate per 100,000 population [12]. Age adjustment removes the primary confounding with ACS demographic variables, which is that states with more old age populations would score much more highly on age-related ACS features. Additionally, the CDC suppresses states with death counts below 10 deaths per state, so small states may be absent depending on the year selected. Cancer incidence is similarly sourced from the CDC's WONDER database for all states, and through years 1999-2022. The cancer types listed span over 100 types of cancers. This data also uses age-adjusted rates per 100,000 population [12].

UFO sighting data is sourced from the National UFO Reporting Center (NUFORC) and covers a massive span of years from the 1800-2016 when reporting changed to a on request basis. I have placed a request for up to date data, and will include the up to date numbers in the analysis below if accepted. As much data as possible will be delivered alongside this project based on distribution limitations imposed by NUFORC.

This dataset is included because it represents a target dataset with no clear causal relationship to any ACS demographic variables. Signals surfaced by the pipeline for this dataset are most likely to show reporting bias. By including this dataset, the pipeline will

demonstrate how results require further decomposition to determine the distinction between correlation and causation.

Recipe Sampler & ML Layer

Recipe sampling design

Once the candidate feature set is reduced in size through the screening process, it can be passed to the recipe sampling layer. For features that are weakly correlated with the target data, but more strongly correlated with complementary features, machine learning is well suited to find these signals. Recipe sampling is the method that will find combinations of these features while maintaining a manageable feature set across tests. A recipe is randomly sampled from the screened candidate set, then trained and scored with a CatBoost model. Over many recipes, the features that appear in high scoring recipes accumulate higher scores for itself and its family.

The sampling has 2 phases: Coverage sampling and Stratified Random sampling. The coverage phase ensures every feature in the candidate set is included in a recipe a configurable amount of times. The stratified random sampling phase prefers sampling features from higher scoring families. This stratified sampling will converge towards families and subfamilies with closer correlation to the target dataset.

CatBoost evaluation per recipe

The recipes produced by the stratified sampling are evaluated by training a CatBoost regression model on only the features included in the recipe. Therefore, the recipe is a test of whether a particular subset of the screened Census variables can predict the target data with useful accuracy. Prior to training, the system pulls the recipe designated feature columns, removes rows with missing data, and coerces the values into numeric form. This guarantees that each recipe is a training dataset with only valid and usable feature data columns.

The CatBoost model is configured using RMSE as the training loss metric and evaluation metric. By shallow trees of depth 3, 300 iterations and learning rate of 0.03 we have a more constrained approach that will optimize for performance. Each recipe is evaluated using 3-fold cross-validation. This means that in every fold the CatBoost model is trained on only 2/3 of the geography rows and gets evaluated on the hold-out 1/3. The model will make predictions for the validation target values, those predictions are compared against the actual target values using R^2, RMSE and MAE [13]. The score metric for R^2 is shown below.

$$R^2 = 1 - \frac{\sum \left(y_i - \hat{y}_i\right)^2}{\sum \left(y_i - \bar{y}\right)^2}$$

In this formula y_i is the actual target value for the geography row i, $\hat{y}_i$is the CatBoost prediction for that geography row, and $\bar{y}$ is the mean actual target data in the validation fold. R^2 score near 1 indicate that the recipe is good at predicting the actual value, while scores near 0 indicate that the model is only good at predicting the average target value. Resulting R^2 scores can be negative, this means that the recipe is actively worse at predicting the target data. Such a result is considered very meaningful, as it shows which features confound predictive analysis of the data.

Score aggregation: column, family, pairwise synergy

For each trained recipe, the validation scores are averaged across three folds. The resulting mean R^2 is the recipe's primary performance score, while RMSE and MAE are kept as error metrics. By averaging results across folds, it prevents a small group of highly influential geography rows from overly affecting the score and ensuring a stable estimate of the recipe's predictive power.

The resulting model recipe score is aggregated among all other recipes. Features that consistently appear in high or low scoring recipes distribute their own scores up the tree of

hierarchy. This attributes scores of features to the subfamilies and families that the feature belongs to.

Features that score better when paired with other features than on average will contribute to their pairwise score. This is an additional scoring metric that helps inform if a feature can actively help or hurt a model's predictive ability, even if by itself it is not a high contributor to predictive ability.

Final model training

Once the recipe sampling has identified the strongest combinations of features, a final training is performed on the highest-scoring recipe as the recommended feature set for the target data. The recipe is selected of the highest mean validation R^2. Due to the model choice of CatBoost, all recipe training iterations are directly comparable with each other, and the recipe that performed best at predicting hold-out target data is selected for the final recipe.

The final model training stage uses the same configuration and data cleaning rules as the recipe evaluation. This ensures that the data is evaluated using the same assumptions as the evaluation that it was originally configured with. The output of this model training is returned to the front-end along with some important metrics about the training. These include: the selected top recipe, aggregated feature-level scores, family-level scores, and pairwise scores.

Pipeline Architecture

The beginning of the pipeline aligns the user's target dataset with ACS geography using the FIPS code for state and county as the shared identifier. Since the user dataset might have incomplete geographic coverage, the geographies missing target values are removed rather than being filled in with zero. This is because missing targets may represent lack of coverage or non-reporting, like in a case where a state does not produce enough quantity of a crop to be reported in USDA data. This produces a feature matrix X and a target vector y which will

be passed to the statistical screening layer. Every ACS geography row is compared with the user target variable using both Pearson and Spearman correlations, and followed by Benjamini-Hochberg FDR correction. If the correlation values are above the minimum thresholds, the feature is considered to be passing. The output of this stage is the statistical screening report, which provides both reporting information about the screening analysis but also a filtered dataset of only features that passed.

The next part of the pipeline reads the screening report to produce a feature set that is ready for machine learning. The pools of stable and passing features as well as feature families are parsed and sampled to produce recipes with different subsets of features from the screened feature set. Recipes are crafted for the selected years of the screened ACS data and user target data. The recipe sampler builds for recipes with a limited number of ACS variables and evaluated using CatBoost using R^2, RMSE, and MAE. As an intermediate step, the evaluation results are sent to the Godot frontend for data visualization charts.

After recipe evaluations are complete, the system aggregates the scores into several layers of outputs. Features scores get calculated using the mean performance of recipes that contain that feature. Family and subfamily scores are calculated as aggregate from their individual member features' scores. Due to this aggregation taking place after the screening step, not all families and subfamilies may be present as a result of being removed. Pairwise scores are also collected from the recipe's containing combinations of paired features compared with the average performance of those features. These score aggregations are collected into a recipe sampling report and returned to the Godot front-end. This report is parsed and data visualizations are built. This allows the user to view feature scores, family scores, and pairwise relationships without inspecting the raw report.

Cross-dataset Comparison

Each of the datasets were analyzed using the pipeline to best understand how the context of datasets is related to the results of the analysis. For the statistical screening phase, the Cancer dataset and UFO sightings dataset have high mean absolute R correlation values. This demonstrates how data that is significantly confounded by population can affect the features that pass the screening phase. By contrast, the Opioid related deaths dataset has a low mean absolute R value of ~0.148. During the screening phase, convergence was not achieved as most ACS features did not have sufficient correlation with the target data.

In the recipe evaluation phase, the Cancer dataset has a very high R2, this indicates that the model is very reliable at predicting the target data using ACS demographic features. The Opioid dataset produces mean R2 near zero with individual recipes at max 0.6. This shows that while most features do not model the signal well, there are real yet sparse signals. By contrast, UFO sightings data starts at mean 0.174 and max 0.8 but at 2016 the signal collapses with mean -0.87 and max 0.042. This is explained with a breakdown in the reporting data availability from the dataset.

Problems & Possible Improvements

During the planning of the statistical screening phase, BH-FDR was considered to maintain stability between the Pearson's and Spearman correlation step. However BH-FDR did not have any effect on any of the features that passed or failed the screening phase. I would consider a different method or to remove BH-FDR altogether from that step.

In the model evaluation phase, it was found that UFO sightings data had the most exciting results. However the analysis broke down at the last year of available data. This is not considered a direct result of lack of signal in feature to target data, but rather a breakdown in the availability of data. In 2017, NUFORC changed the terms by which their full dataset would

be provided. After that year, it was required that an individual reach out to them and get permission specifically to access and make the data available even for research purposes. Contacting NUFORC for the full up to date dataset would have provided tremendous context to what was originally thought to be a stress-testing dataset but was actually one of the most meaningful sources of data in the project. I contacted their team but did not receive a response in time to include it in this report.

Lastly, alternative machine learning models would be worth testing in the system. This would require significant normalization of the user target data, and additionally would require strict full coverage data as small gaps in available data would cause significant problems for other datasets.

Bibliography

1. “What Is Data.Census.Gov?” *Census.Gov*, US Census Bureau, 17 Dec. 2025, www.census.gov/data/what-is-data-census-gov.html.
2. “Decennial Census of Population and Housing.” *Census.Gov*, US Census Bureau, 25 May 2024, www.census.gov/programs-surveys/decennial-census.html.
3. US Census Bureau. “The 72-Year Rule.” *Census.Gov*, 3 Sept. 2024, www.census.gov/about/history/census-records-family-history/census-records/72-year-rule.html.
4. US Census Bureau. “The History of Privacy and Confidentiality at the U.S. Census Bureau.” *Census.Gov*, 26 Feb. 2024, www.census.gov/about/history/bureau-history/agency-history-timeline/privacy-confidentiality.html.
5. GeeksforGeeks. “Principal Component Analysis (PCA).” *GeeksforGeeks*, 13 Nov. 2025, www.geeksforgeeks.org/data-analysis/principal-component-analysis-pca/.
6. “K- Fold Cross Validation in Machine Learning.” *GeeksforGeeks*, GeeksforGeeks, 12 Sept. 2025, www.geeksforgeeks.org/machine-learning/k-fold-cross-validation-in-machine-learning/.
7. Prokhorenkova, Liudmila, et al. “CatBoost: Unbiased Boosting with Categorical Features.” *NeurIPS*, Moscow Institute of Physics and Technology, 28 June 2017, papers.neurips.cc/paper/7898-catboost-unbiased-boosting-with-categorical-features.pdf.

8. Frost, Jim. “Pearson Correlation Coefficient.” Statistics By Jim, www.statisticsbyjim.com/basics/correlations/

9. Roberts, Michelle R., Sepideh Ashrafzadeh, and Maryam M. Asgari. “Research Techniques Made Simple: Interpreting Measures of Association in Clinical Research.” Journal of Investigative Dermatology, vol. 139, no. 3, Mar. 2019, pp. 502–511.e1, doi:10.1016/j.jid.2018.12.023.

10. Benjamini, Yoav, and Yosef Hochberg. “Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing.” Journal of the Royal Statistical Society: Series B (Methodological), vol. 57, no. 1, 1995, pp. 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x.

11. Robinson, W. S. “Ecological Correlations and the Behavior of Individuals.” American Sociological Review, vol. 15, no. 3, 1950, pp. 351–357. doi:10.2307/2087176.

12. Centers for Disease Control and Prevention. "Age-Adjusted Rates." CDC WONDER Multiple Cause of Death Help, wonder.cdc.gov/wonder/help/mcd-expanded.html#Age-Adjusted%20Rates.

13. Pedregosa, Fabian, et al. "Scikit-learn: Machine Learning in Python." Journal of Machine Learning Research, vol. 12, 2011, pp. 2825–2830, jmlr.org/papers/v12/pedregosa11a.html.